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PROFESSIONAL SUMMARY

AI Engineer specializing in Machine Learning, Generative AI, Computer Vision, and intelligent automation, with hands-on experience designing, developing, and deploying end-to-end AI applications. Experienced in building ML pipelines with Python, PyTorch, TensorFlow, and Scikit-learn, developing LLM-powered systems using OpenAI APIs and function calling, and deploying AI services with FastAPI. Passionate about building reliable AI solutions that solve real-world problems.

EXPERIENCE

AI Engineering Intern (Agentic AI Developer Track) Jul 2025 – Present
National Telecommunication Institute (NTI) – Digital Egypt Youth (DEY) HireReady Initiative

- Selected for a competitive 420-hour track focused on building autonomous AI agents and LLM-powered applications.
- Applying LangChain, LangGraph, and OpenAI APIs to develop agentic AI workflows and tool integration.
- Working with advanced AI engineering concepts including prompt engineering, AI safety, deployment, scaling, and monitoring.
- Collaborating with mentors and peers on AI engineering projects following industry best practices.

SELECTED PROJECTS

StructuraNet AI – AI-Powered Network Design (Graduation Project, Full Mark) GitHub
Role: AI Engineer – Python AI Engine, LLM Orchestration & AI Automation (8-member team)

- Defined the end-to-end AI architecture, separating probabilistic LLM reasoning from deterministic Python execution to improve reliability, maintainability, and validation.
- Architected a Python AI engine that transforms natural-language network requirements into validated, deployment-ready GNS3 projects.
- Engineered an LLM orchestration pipeline using OpenAI APIs and function calling, eliminating rigid finite-state machines and enabling dynamic multi-step reasoning.
- Built a layered validation framework: deterministic auto-repair of LLM output, feedback-guided retries, a three-gate safe merge that prevents LLM-generated configurations from modifying hardware properties, and an 11-check GNS3 structural validator.
- Integrated the Python AI engine with cross-functional backend (Node.js) and frontend (React) systems via a persistent JSONL bridge to deliver a complete AI solution.
- Automated the generation of enterprise-grade Cisco IOS security configurations through configurable security profiles.

Brain Tumor Classification from MRI – Medical AI GitHub

- Designed and trained a CNN from scratch in **PyTorch** for four-class brain MRI classification (No Tumor, Pituitary, Meningioma, Glioma), reaching **97% accuracy** on 1,311 held-out test scans (per-class ROC-AUC ≥ 0.996).
- Performed data preprocessing, model evaluation, and hyperparameter optimization, using dropout regularization to improve generalization.
- Integrated the trained model into a Hospital Management System deployed on **Microsoft Azure** as the DEPI Data Engineering final project at ITC College.

Trash Classification using Transfer Learning – Computer Vision GitHub

- Engineered a transfer-learning image classification pipeline using **TensorFlow** and MobileNet for six-class waste recognition.
- Trained with heavy data augmentation, learning-rate scheduling, and early stopping, reaching **85% validation accuracy** (macro F1 0.84) on 508 held-out images.
- Evaluated performance using confusion matrix, precision, recall, F1-score, and per-class classification reports.

Football Player Detection – Object Detection GitHub

- Fine-tuned **YOLOv5x** for 100 epochs on a Roboflow dataset to detect players, goalkeepers, referees, and the ball, and ran inference on match video at about 30 ms per frame.

Diabetes Prediction API – ML Deployment GitHub

- Developed and deployed a containerized **FastAPI** inference service exposing a Scikit-learn model through REST APIs, with a median response time of about 2 ms in a 500-request local benchmark.
- Built an end-to-end deployment pipeline using Scikit-learn, Pickle, **Docker**, and FastAPI, and published the image to Docker Hub.

Car Price Prediction Web App – ML Regression GitHub

- Benchmarked four regressors and selected Random Forest on a used-car dataset, achieving an RMSE of 5,169 and R^2 of 0.80 on the test set.
- Deployed a **Streamlit** web application for real-time price prediction based on user inputs.

TECHNICAL TRAINING

Generative AI Professional – DEPI – AMIT Jan 2025 – Jun 2026

- Advanced professional training covering LLMs, prompt engineering, AI agents, OpenAI APIs, and production-oriented Generative AI workflows.

Deep Learning for Computer Vision – NTI (72 Hours)

- Built image classification and object detection models using Python, OpenCV, and TensorFlow.
- Applied preprocessing, segmentation, and YOLO-based detection workflows.

Data Engineering Bootcamp – DEPI (6 Months)

- Designed ETL pipelines and gained hands-on experience with Microsoft Data Engineering tools.

Machine Learning Training – NTI (72 Hours)

- Trained predictive ML models achieving up to 90% accuracy on various classification tasks.

CERTIFICATIONS

- HCIA-AI V4.0 Course – Huawei (Jan 2026).
- Python Project for Data Engineering – Coursera (Apr 2026).
- Web Scraping with Scrapy & Python – Packt (Apr 2026).
- Generative AI: Learn about the next AI frontier – CoursatAI (Nov 2025).
- Deep Learning for Natural Language Processing (NLP) – Udemy (Oct 2025).
- Microsoft Data Engineer – Digital Egypt Pioneers Initiative (DEPI).
- Machine Learning Level 1 – NTI.

TECHNICAL SKILLS

Programming Languages: Python.
Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, Supervised Learning, Unsupervised Learning, Deep Learning.
LLMs & Agents: OpenAI API, Function Calling, LangChain, LangGraph, Prompt Engineering.
Computer Vision: CNNs, Transfer Learning, YOLO, OpenCV.
Natural Language Processing: Tokenization, Embeddings, Transformers, RNN, LSTM, Hugging Face.
Deployment & Cloud: FastAPI, REST APIs, Streamlit, Docker.
Data & Utilities: Pandas, NumPy, Matplotlib, Seaborn, Scrapy.

EDUCATION

B.Sc. Computer Engineering and Automatic Control 2021 – 2026

Faculty of Engineering, Tanta University / Grade: Excellent

- Relevant AI Coursework: Neural Networks, Artificial Intelligence, Digital Image Processing, Data Structures, Object-Oriented Programming, Control Systems, Microprocessors, Digital Logic Design.